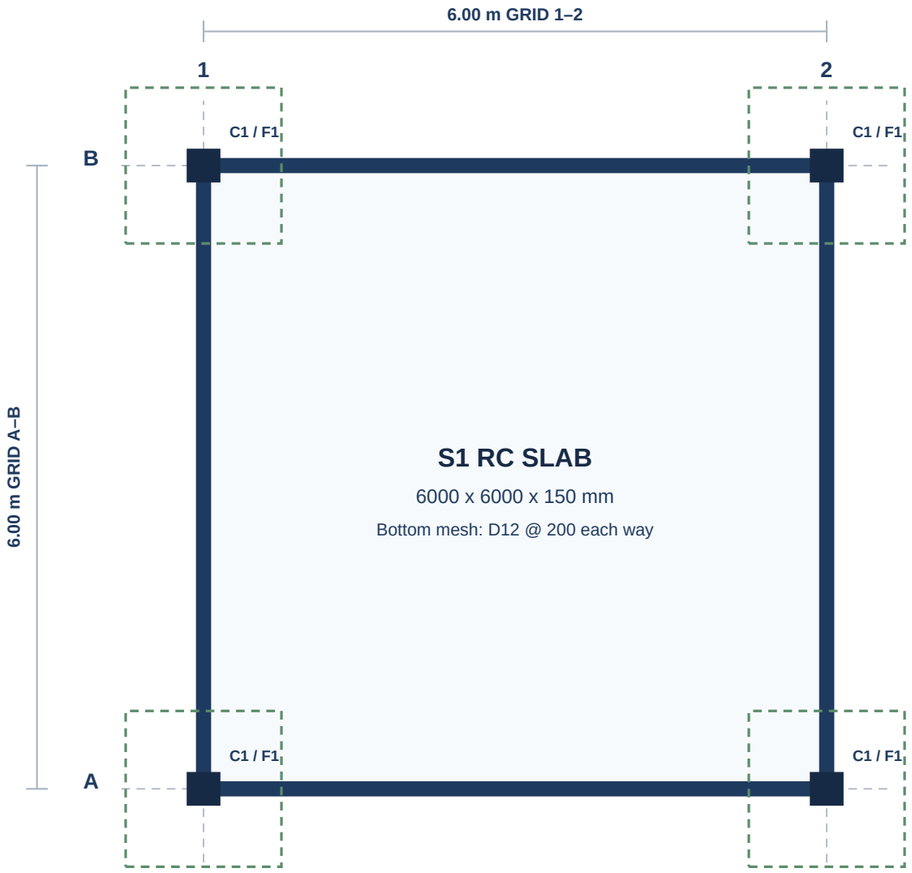
MMC AI | SYNTHETIC STRUCTURAL TAKEOFF TEST

FOUNDATION AND FRAMING PLAN

TEST ONLY — NOT FOR CONSTRUCTION



STRUCTURAL ELEMENT SCHEDULE

F1 — Isolated footing	4 No. | 1.50 x 1.50 x 0.45 m
C1 — Reinforced column	4 No. | 0.30 x 0.30 m | height 3.00 m
B1 — Perimeter beam	4 No. | 250 x 450 mm overall
B1 clear beam length	5.70 m between column faces
S1 — RC slab	1 No. | 6.00 x 6.00 x 0.15 m

REFERENCE DIMENSIONS

- Plan grid: 6.00 m x 6.00 m between column axes.
- Columns are 300 x 300 mm; footings are 1500 x 1500 x 450 mm.
- Beam section is 250 x 450 mm overall; slab is 150 mm thick.
- For concrete, count slab thickness over the full 6 x 6 panel.
- Count beam concrete below slab only: 250 x (450-150) mm.
- Beam clear length is 5.70 m; no beam/column overlap is counted.
- Column height is 3.00 m measured above footing top.

Synthetic test geometry; no code design or construction use is implied.

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REINFORCEMENT DETAILS AND BAR SCHEDULE

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F1 — FOOTING REINFORCEMENT

4 identical pad footings; each 1500 x 1500 x 450 mm.

Bottom reinforcement: D12 at maximum 150 mm spacing, each way.

Concrete cover: 50 mm. Bar cut length: 1.40 m.

Per footing: 11 bars X + 11 bars Y; 44 bars per direction for all 4 footings.

No top bars, laps or wastage included in the test schedule.

C1 — COLUMN REINFORCEMENT

4 identical columns; section 300 x 300 mm; height 3.00 m.

Longitudinal steel: 8-D16 per column; scheduled cut length 3.60 m.

Ties: D8 at maximum 150 mm spacing; 21 ties per column.

Tie scheduled cut length: 1.10 m, including hooks.

No lap allowance or wastage included.

B1 — PERIMETER BEAM REINFORCEMENT

4 beams; 250 x 450 mm overall; clear length 5.70 m each.

Slab thickness is 150 mm; beam concrete below slab is 300 mm deep.

Longitudinal steel: 2-D16 top + 2-D16 bottom; cut length 6.40 m each.

Stirrups: D8; 41 per beam; scheduled cut length 1.20 m each.

S1 — SLAB REINFORCEMENT

One 6000 x 6000 x 150 mm slab panel.

Bottom mesh: D12 at maximum 200 mm spacing, each way.

Concrete cover: 50 mm; scheduled bar cut length 5.90 m.

31 bars in X + 31 bars in Y; laps and wastage excluded.

BAR SCHEDULE — SCHEDULED CUT LENGTHS ARE AUTHORITATIVE

MARK	ELEMENT / DESCRIPTION	BAR	NO.	CUT LENGTH	TOTAL LENGTH
F1-D12-X	Footings, bottom bars, X direction	D12	44	1.40 m	61.60 m
F1-D12-Y	Footings, bottom bars, Y direction	D12	44	1.40 m	61.60 m
C1-D16-MAIN	Column longitudinal bars, 8 per column	D16	32	3.60 m	115.20 m
C1-D8-TIES	Column ties, 21 per column	D8	84	1.10 m	92.40 m
B1-D16-MAIN	Beam longitudinal bars, 4 per beam	D16	16	6.40 m	102.40 m
B1-D8-STIRRUPS	Beam stirrups, 41 per beam	D8	164	1.20 m	196.80 m
S1-D12-X	Slab bottom mesh, X direction, 31 bars	D12	31	5.90 m	182.90 m
S1-D12-Y	Slab bottom mesh, Y direction, 31 bars	D12	31	5.90 m	182.90 m

Theoretical steel mass convention for answer key: kg/m = diameter(mm)^2 / 162. No laps, chairs, spacers, couplers, or waste.

Drawing is synthetic and for software testing only; it is not a design, permit, or construction document.